



EXPERIMENT ON VERIFICATION OF KIRCHHOFF'S VOLTAGE LAW (KVL)

PROFESSOR AKINOLA OLAYINKA,
PhD Physics



AIM

Verification of Kirchhoff's Voltage law.



LEARNING OBJECTIVES

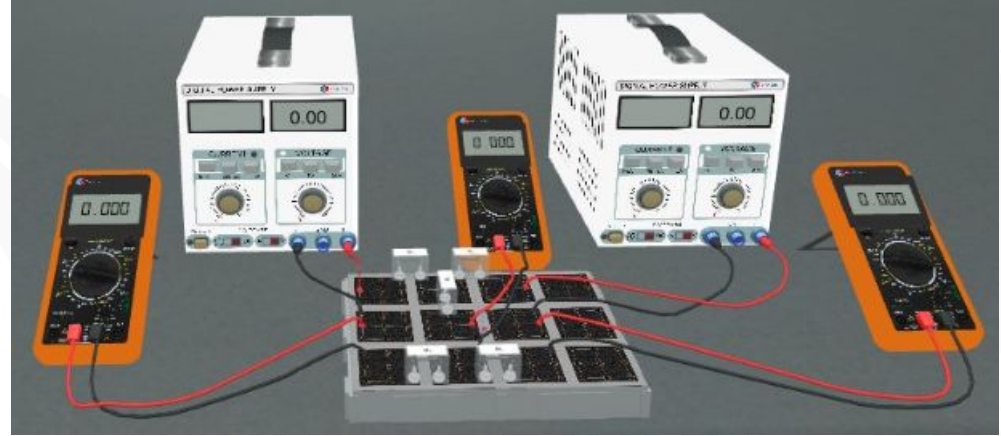
At the end of this lesson, you should be able to:

- State Kirchhoff's Current law (KCL) for electric circuits.
- Apply Kirchhoff's current law at node (or junction) points in an electric circuit.
- Apply Kirchhoff's voltage law around closed path within electric circuits.
- Deduce the values of the currents in different branches of an electric circuit.



APPARATUS

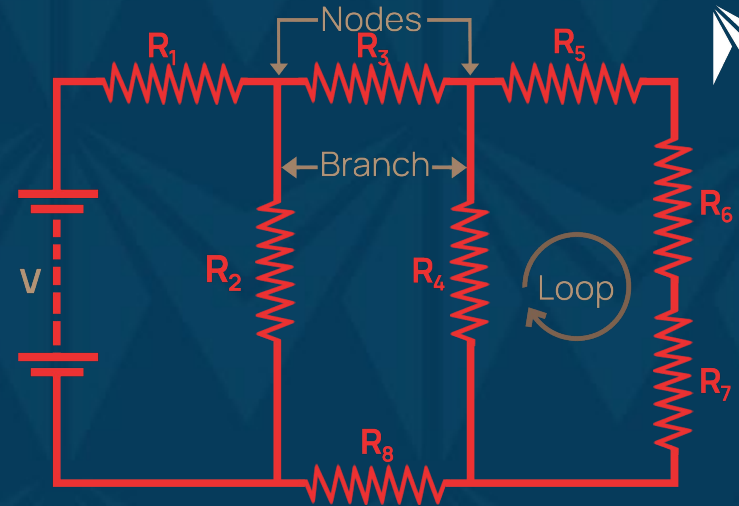
- Bread board.
- Two DC power supplies.
- Five variable resistors.
- Three ammeters.
- Connection cables.





THEORY

- A node is a point or junction in an electrical circuit where two or more components, such as wires or electrical devices, meet.
- A branch in an electrical circuit is a single path that connects two nodes or junctions. It typically consists of one or more electrical components, such as resistors, capacitors, or other circuit elements, arranged in series or parallel.
- A loop, often referred to as a closed loop or mesh, is a closed pathway within an electrical circuit formed by interconnected branches.





THEORY

Kirchhoff's Voltage Law (KVL)

It is based on the principle of conservation of energy.

KVL states that around any closed path, the sum of all voltage drops on all branches within the loop is equal to the sum of the emf's of the batteries within the loop.

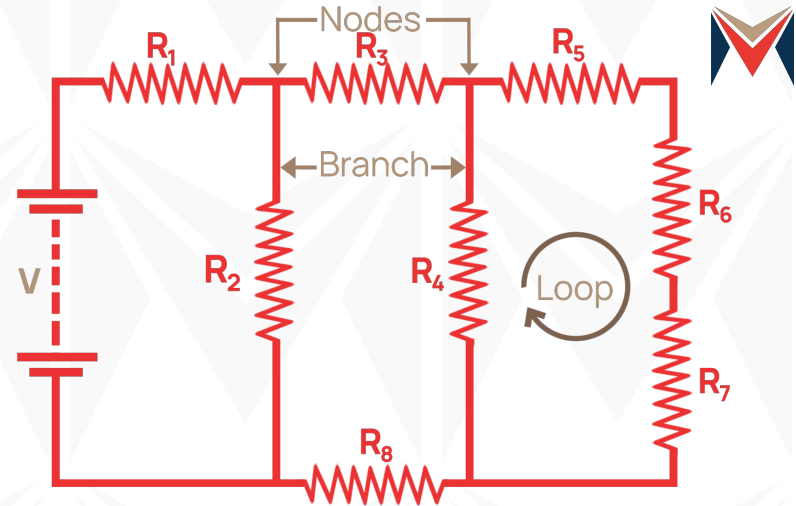
Or

Around any closed loop or path in a circuit, the algebraic sum of all the voltage drops must equal zero.

$$\sum V = \sum \varepsilon$$

or

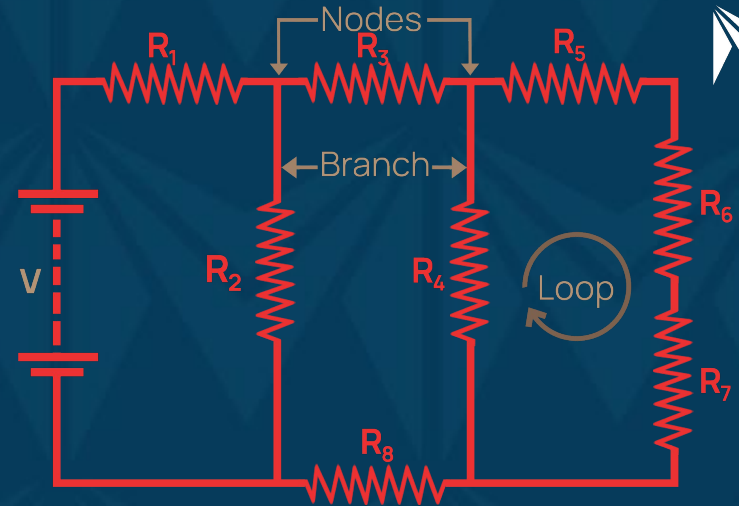
$$\sum V = 0$$





THEORY

- The potential difference is considered positive, if the resistance is traversed in the same direction as the current flowing through it and negative otherwise.
- The emf of the battery is considered positive if the battery is traversed in the circuit from positive terminal to its negative terminal in the loop and negative otherwise.

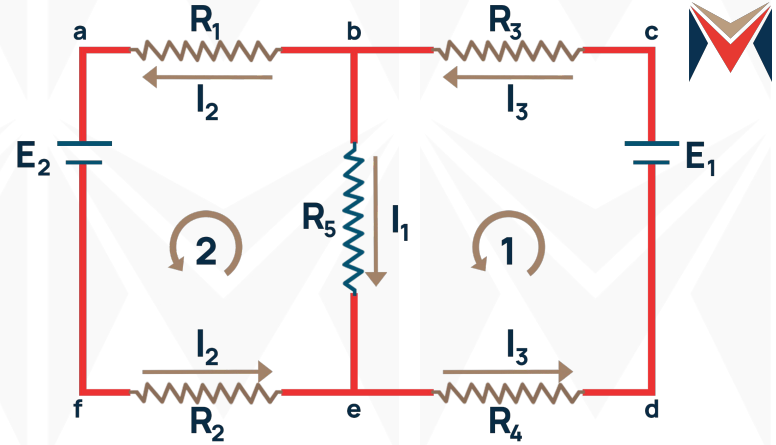




THEORY

- Apply KCL to node b in the figure,
 $I_1 + I_2 = I_3$ (1)
- Apply KVL to loop 1 in the figure,
 $I_3 R_3 + I_3 R_4 + I_1 R_5 = E_1$ (2)
- Apply KCL to node b in the figure,
 $I_2 R_2 - I_1 R_5 + I_2 R_1 = -E_2$ (3)

By solving the above equations
(analytically), we get the values of I_1 , I_2 , and
 I_3 .





MID-LESSON QUESTIONS

Question :

Q1: State Kirchhoff's Voltage Law (KVL)

Q2: In a simple circuit with two resistors in series and a battery as the voltage source, how does KVL apply when you follow a loop around the circuit?



MID-LESSON ANSWERS

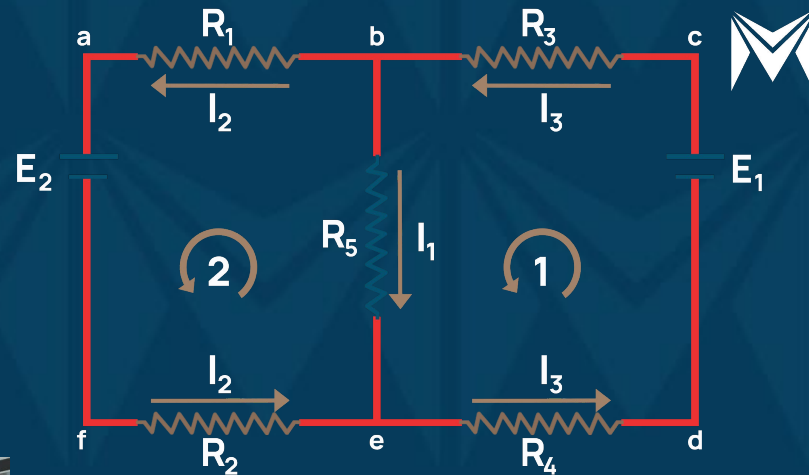
A1: Kirchhoff's Voltage Law (KVL) states that the sum of the voltages (emf or voltage sources) and voltage drops within any closed loop or mesh in an electrical circuit is equal to zero.

A2: When you follow a loop around the circuit, KVL tells us that the sum of the voltage drops across the resistors must be equal to the voltage supplied by the battery. Mathematically, this can be expressed as $V_{\text{battery}} = V_{R1} + V_{R2}$, where V_{battery} is the battery voltage, and V_{R1} and V_{R2} are the voltage drops across the resistors.



PROCEDURE

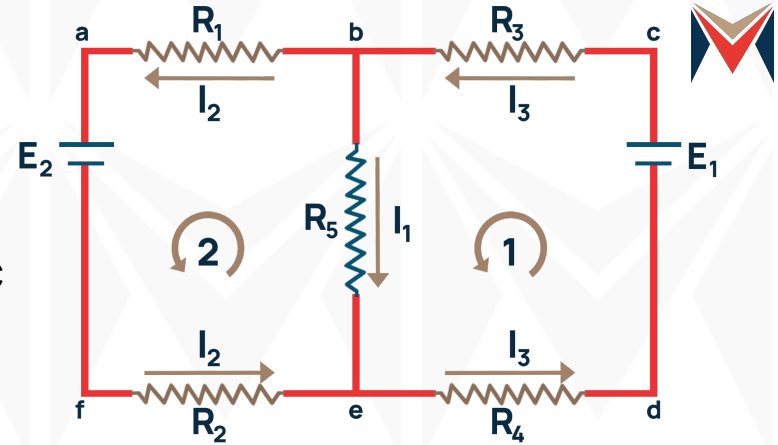
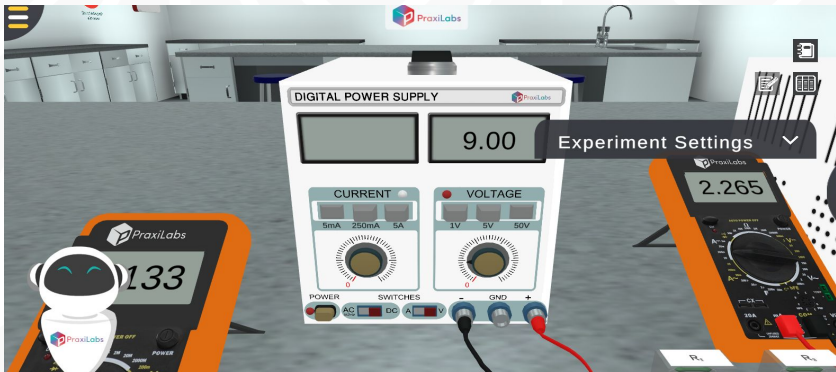
1. Select the values of R_1 through R_5 and connect the circuit as shown.





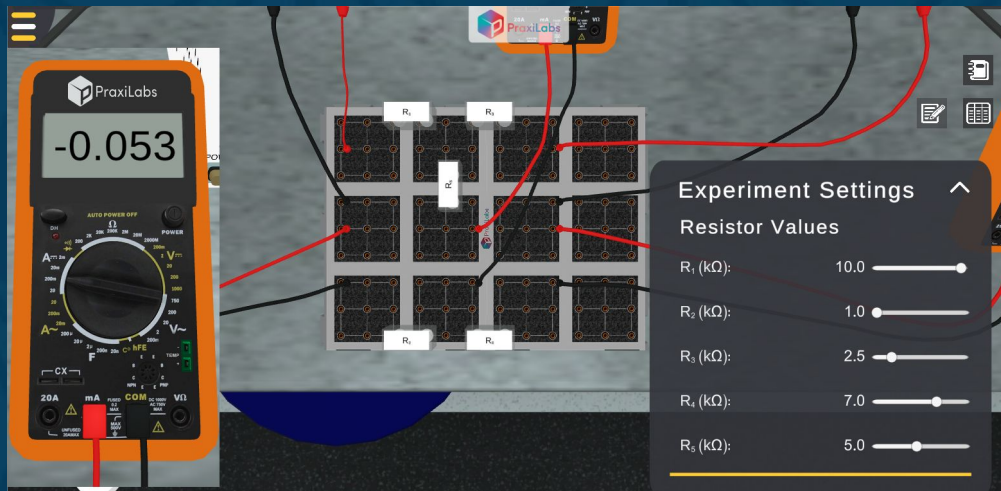
PROCEDURE

2. Select the maximum output voltage of both DC power supplies to be 50 V by pressing the 50 V button above the voltage dial.
3. Adjust the emf's of both supplies to 9V using the voltage dial knob.

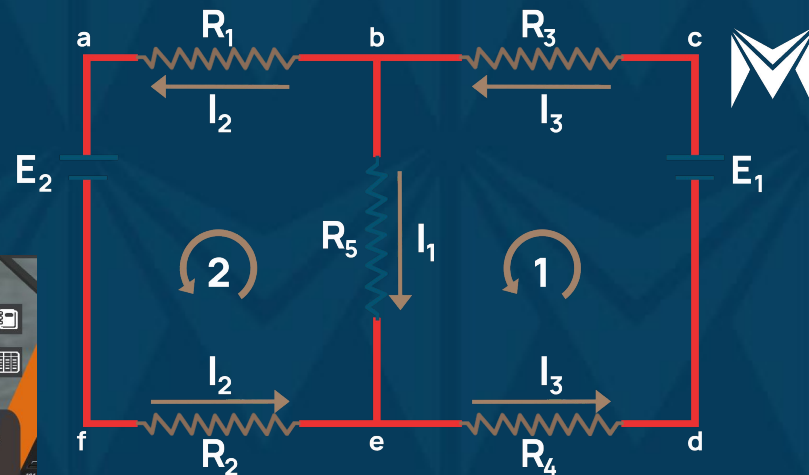




PROCEDURE



4. Click the Record button to record the readings of the three ammeters.
5. Increase the EMF E_2 at regular intervals (3V) and repeat step 4





PROCEDURE

R_1 (k Ω)	R_2 (k Ω)	R_3 (k Ω)	R_4 (k Ω)	R_5 (k Ω)	E_2 (V)	E_1 (V)	I_1 (mA)	I_2 (mA)	I_3 (mA)	$I_{3\text{-computed}}$ (mA)
10	1	2.5	7	5	5	9	9	0.895	-0.416	0.479
10	1	2.5	7	5	5	9	12	1.055	-0.344	0.711
10	1	2.5	7	5	5	9	15	1.215	-0.271	0.944

6. Substitute the values of the resistances and emf's used in Equations 1, 2, and 3.
7. Solve the equations for I_1 , I_2 , and I_3 .
8. Repeat the above steps for different values of the resistances and the emf's of the DC power supplies.
9. Compare between the values calculated and the experimental values and then comment on your results.



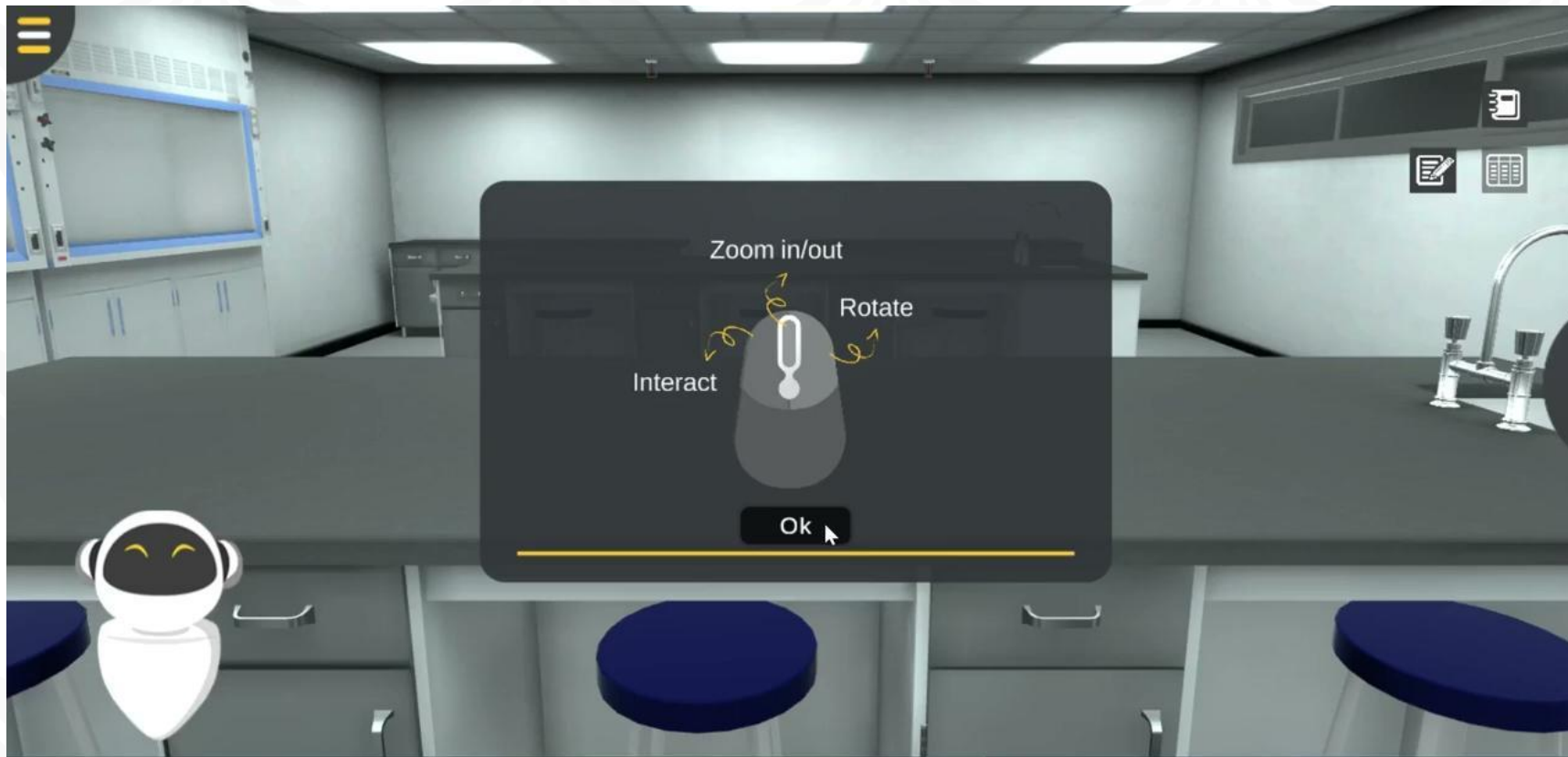
DATA ANALYSIS

R_1 (k Ω)	R_2 (k Ω)	R_3 (k Ω)	R_4 (k Ω)	R_5 (k Ω)	E_2 (V)	E_1 (V)	I_1 (mA)	I_2 (mA)	I_3 (mA)	$I_{3\text{-compute}}$ I_d (mA)
10	1	2.5	7	5	9	9	0.895	-0.416	0.479	
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- Using the KCL and KVL equations (1) to (3)
- Calculate the I_1 , I_2 and I_3
- Compare between the values calculated and the experimental values of I_3 and then comment on your results.



WALK_THROUGH VIDEO





SUMMARY

In this lesson, we explored Kirchhoff's Voltage Law (KVL), a fundamental principle in electrical circuit theory, emphasizing that the sum of voltages in a closed loop equals zero due to the conservation of energy. We discussed the importance of experimentally verifying KVL to ensure its real-world applicability. The experimental procedure involves setting up a circuit, measuring voltages, and rigorously analyzing data. This verification process bolsters confidence in KVL's accuracy for practical circuit analysis and design.



FURTHER READING

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THANK
YOU